

# Triangles: their kinds and their elements

Three sides, three angles, and the two ways of classifying every triangle there is.

LESSON 46

1 × 40 MIN

MATEMATIKA 7, §19

S8 8.1

LESSON CLOCK

10'

12'

12'

6'

|                         |        |
|-------------------------|--------|
| Elements and notation   | 10 min |
| Two classifications     | 12 min |
| The triangle inequality | 12 min |
| Homework                | 6 min  |

LEARNING OBJECTIVES

- Name the elements of a triangle and use the standard notation.
- Classify triangles by their sides and by their angles.
- Use the triangle inequality to decide whether three lengths can form a triangle.
- Identify the median, the altitude and the bisector of a triangle.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 118–122        |
| Cambridge | Stage 8, pp. 78–84 |
| Workbook  | Exercise 8.1       |

01

## Explanation

### Elements and notation

A triangle has three **vertices**, three **sides** and three **angles**. The standard notation names the side by the small letter of the opposite vertex.

| ELEMENT   | WRITTEN                        |
|-----------|--------------------------------|
| vertices  | $A, B, C$                      |
| sides     | $a = BC, b = CA, c = AB$       |
| angles    | $\angle A, \angle B, \angle C$ |
| perimeter | $P = a + b + c$                |

Three special segments from a vertex:

| SEGMENT  | GOES TO                              |
|----------|--------------------------------------|
| median   | the midpoint of the opposite side    |
| altitude | the opposite side, at a right angle  |
| bisector | the opposite side, halving the angle |

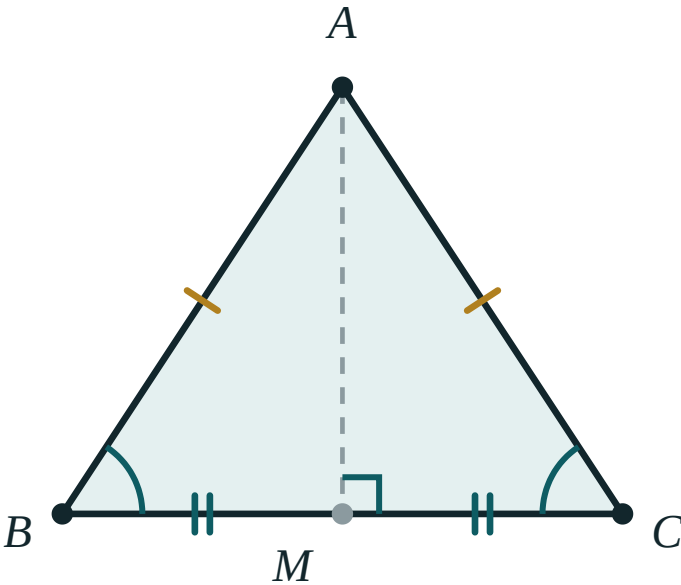
SIDE A FACES ANGLE A

That single convention makes every later formula readable — the sine rule, the cosine rule, the area formula. It is worth fixing now.

Two classifications

| BY SIDES    | MEANS               | BY ANGLES | MEANS            |
|-------------|---------------------|-----------|------------------|
| scalene     | all sides different | acute     | all angles acute |
| isosceles   | two sides equal     | right     | one angle 90°    |
| equilateral | all three equal     | obtuse    | one angle obtuse |

The two classifications are independent: a triangle may be isosceles and right, or scalene and obtuse, and so on. An equilateral triangle is always acute, with all angles 60°.



$AB = AC \implies \angle B = \angle C$

An isosceles triangle — two equal sides and two equal base angles.

EVERY EQUILATERAL TRIANGLE IS ISOSCELES

“Two sides equal” does not mean “exactly two”. The equilateral triangle is the special case in which all three are.

The triangle inequality

Each side is less than the sum of the other two

$$a < b + c, b < c + a, c < a + b$$

It is enough to check the **longest** side: if the longest is less than the sum of the other two, all three inequalities hold.

| LENGTHS   | CHECK         | TRIANGLE? |
|-----------|---------------|-----------|
| 3, 4, 5   | $5 < 3 + 4$   | yes       |
| 2, 3, 6   | $6 > 2 + 3$   | no        |
| 4, 4, 8   | $8 = 4 + 4$   | no — flat |
| 7, 10, 15 | $15 < 7 + 10$ | yes       |

EQUALITY GIVES NO TRIANGLE

4, 4, 8 lays the two short sides flat along the long one. The inequality must be strict.

**02** Worked examples**EXAMPLE 1** Can a triangle have sides 2, 3, 6? And 7, 10, 15?

- ① Longest of the first: 6.  
-----
- ②  $2 + 3 = 5 < 6$  — no triangle.  
-----
- ③ Longest of the second: 15.  
-----
- ④  $7 + 10 = 17 > 15$  — yes.

**ANSWER**

No; yes

**EXAMPLE 2** A triangle has  $a = 5$  and  $b = 9$ . Between which values must  $c$  lie?

- ①  $c < 5 + 9 = 14$   
-----
- ②  $9 < 5 + c \Rightarrow c > 4$   
-----
- ③ Combine.  
-----
- ④  $4 < c < 14$

**ANSWER** $4 < c < 14$

**EXAMPLE 3** Classify a triangle with sides 5, 5, 8 and one angle of  $106^\circ$ .

- ① Two sides equal: isosceles.  
By sides.
- ② One angle above  $90^\circ$ : obtuse.  
By angles.
- ③ Both classifications apply at once.
- ④ An obtuse isosceles triangle.

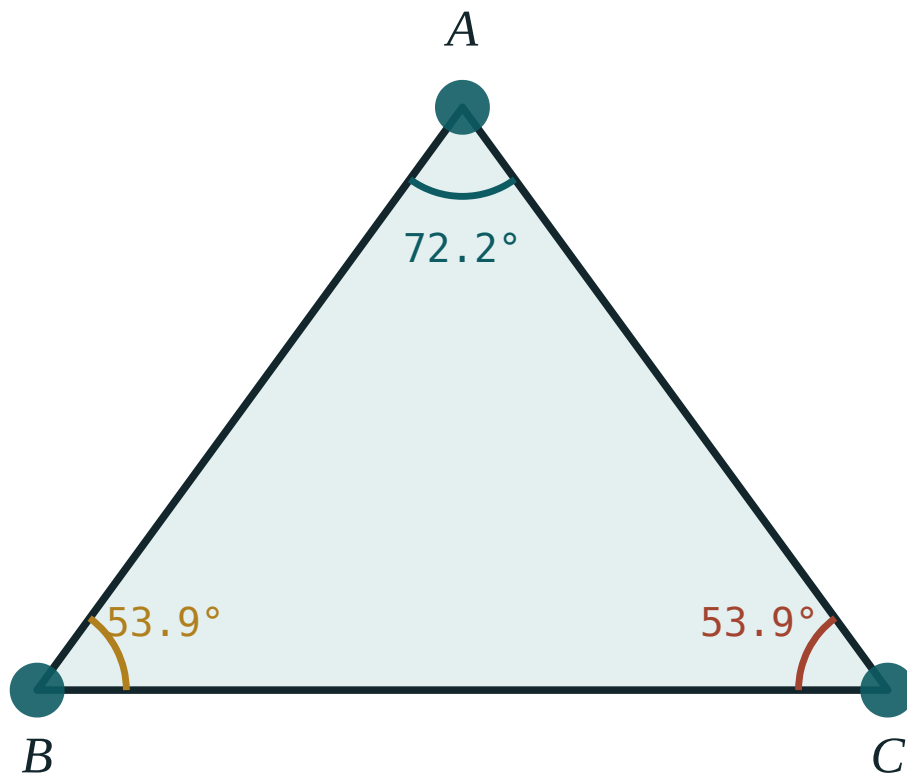
**ANSWER**

Obtuse isosceles

**03 Interactive model**

Give out three strips of card of lengths 2, 3 and 6 cm and ask the class to make a triangle; the failure is more convincing than the inequality.

## Kinds of triangle

 $\angle A$   $72.2^\circ$  $\angle B$   $53.9^\circ$  $\angle C$   $53.9^\circ$ sum  $180^\circ$ longest side **BC**largest angle  $\angle A$ 

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH     | O‘ZBEKCHA     | РУССКИЙ        |
|-------------|---------------|----------------|
| Triangle    | Uchburchak    | Треугольник    |
| Vertex      | Uchi          | Вершина        |
| Side        | Tomon         | Сторона        |
| Scalene     | Turli tomonli | Разносторонний |
| Isosceles   | Teng yonli    | Равнобедренный |
| Equilateral | Teng tomonli  | Равносторонний |
| Median      | Mediana       | Медиана        |
| Altitude    | Balandlik     | Высота         |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Side  $a$  is opposite:

A triangle with two equal sides is:

An equilateral triangle has angles:



Can 2, 3, 6 form a triangle?

yes

no

sometimes

only if right

A median goes to:

the midpoint

at a right angle

the bisected angle

the vertex

With  $a = 5$ ,  $b = 9$ ,  $c$  satisfies:

$c < 14$

$c > 4$

$4 < c < 14$

$c = 14$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Side opposite  $\angle B$

ANSWER  $b$

2. A triangle with all sides different

ANSWER Scalene

3. A triangle with all sides equal

ANSWER Equilateral

4. Angles of an equilateral triangle

ANSWER  $60^\circ$  each

5. Can 3, 4, 5 be a triangle?

ANSWER Yes

6. Can 2, 3, 6 be a triangle?

ANSWER No

7. Can 4, 4, 8 be a triangle?

ANSWER No

Medium · the standard question

7 PROBLEMS

1. Can 7, 10, 15 be a triangle?

ANSWER Yes

2.  $a = 5$ ,  $b = 9$ : range of  $c$

ANSWER  $4 < c < 14$

3.  $a = 6$ ,  $b = 6$ : range of  $c$

ANSWER  $0 < c < 12$

4. Classify 5, 5, 8 with a  $106^\circ$  angle

ANSWER Obtuse isosceles

5. Classify 3, 4, 5

ANSWER Right scalene

6. Perimeter of a triangle 7, 9, 12

ANSWER 28

7. A median goes to

ANSWER The midpoint

**Hard · stretch and reason**

7 PROBLEMS

1. An isosceles triangle with perimeter 20 and base 8

ANSWER Sides 6, 6, 8

2. The possible integer values of  $c$  with  $a = 5$ ,  $b = 9$

ANSWER 5 to 13

3. How many of them?

ANSWER 9

4. An isosceles triangle with sides 3 and 7

ANSWER Only 3, 7, 7

5. Why not 3, 3, 7?

ANSWER  $3 + 3 < 7$

6. A triangle with perimeter 12 and integer sides: how many?

ANSWER 3

7. Name them

ANSWER 2,5,5; 3,4,5; 4,4,4

## 07 Homework

### Homework — 5 tasks

Check the longest side against the sum of the other two, every time.

- Can a triangle have sides 4, 6, 11? And 8, 9, 16?
- A triangle has  $a = 7$  and  $b = 11$ . Give the range of  $c$ .
- Classify a triangle with sides 6, 6, 6.
- Classify a triangle with a  $90^\circ$  angle and two equal sides.
- An isosceles triangle has perimeter 26 and base 10. Find the equal sides.